Topic -2 & 3

Class & object:-

class:-

A class is a blueprint (or) template used to create objects. It defines the Properties (data) and behaviors (methods) that the objects created from it will have

* A class is like a design (or) plan.

Architectural blueprint → used to build many houses (by the same plan)

class → used to create many object.

Real world Example

Student

Data (Attributes)	(Methods) Actions
* Name	* study()
* Age	* attendence()
* Pin number	* write_exam()

Instead of writing this information seperately for every student, we can create a "class student" that defines properties & behaviors.

Then we create many student objects from that class.

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Class - Syntax

```
class <classname>:

    # attributes
    # methods
```

Class → blueprint

Object → real instance created from blueprint

Ex

```
class Dog:
    def Bark(self):
        print("Dog is barking")

    def Sleep(self):
        print("Dog is sleeping")
```

★ Data + functions together

- → A class is a blueprint for creation object
- → we can only access the data in the class by the objects -
- → It defines
  - attributes (data)
  - methods (functions)
- → A class itself does not perform actions until objects are created.
- → Python uses the keyword → class

Object:-

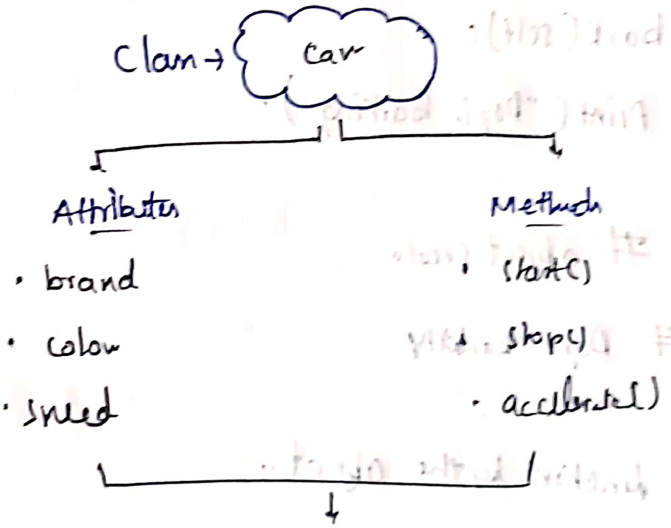
An object is an instance (reference) of a class..

* It is an real entity created from the class blueprint that contains actual data and can perform actions defined in the class.

* A class only defines the structure

"An object is the actual usable thing created from that structure & memory allocated to it"

class → blueprint
object → real instance created from the blueprint



Car1 → Toyota() → object
[Brand = "Toyota"
color = "Red"
speed = 100 K/hr

Car 2 → BMW()

[

)

* The class defines the structure, while each object holds its own data..

~~Exam~~ Syntax

```
object_name = clanName()
```

** Memory allocation takes place to the object itself...

* By object we can call methods from the class.

Ex

```
Class Dog:

    def bark(self):
        print("Dog is barking").

dog1: Dog() # object creation

dog1.bark() # Dog is barking
```

Calling function by the object..

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